

DSE 6311: Data Science Capstone

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Course Term: Summer Session 2

Synchronous Live Session: Wednesdays, 8:00 to 9:30 PM ET

Office Hours: by appointment

Course Description

The data science capstone project is the culminating experience of the data science program, where students work on a data science problem, applying the skills and knowledge they have acquired throughout their studies.

Credit Justification Statement

In this course, students will be expected to do the following amount of work:

- Live class sessions – 90 mins per week 12 hours
- Project proposal 16 hours
- Data acquisition and exploration report 24 hours
- Data preprocessing and feature engineering report 24 hours
- Model selection and development report 24 hours
- Model evaluation and interpretation report 24 hours
- Capstone project report 48 hours
- Peer Review 8 hours
- **Total: 180 hours across the 8-week term**

Course Learning Objectives

Upon completion of this course, students should have the ability to:

- Apply data science methodologies to solve a complex, real-world problem.
- Collect, clean, and preprocess data for analysis.
- Develop and evaluate machine learning models for predictive or classification tasks.
- Communicate findings and insights through reports and presentations.
- Work effectively in a team, if applicable, to complete a collaborative project.

Important Deadlines

Monday 07/07/2025	First day of classes, Summer 2 session
Friday 07/14/2025	Last day to add/drop a course
Monday 08/18/2025	Last day to withdraw from a course with a “W”
Thursday 08/28/2025	Last day of classes
Tuesday 09/02/2025	Final grades due

Required Course Materials

As required for the chosen project.

Required Software:

R & RStudio or Python

Grading & Assignments

A Excellent	B Good	C Fair/Poor	F Unacceptable
93+ A	87-89 B+	77-79 C+	Below 70
90-92 A-	83-86 B	73-76 C	
	80-82 B-	70-72 C-	

Although the graduate grading policy is similar to the undergraduate policy, it should be noted that the expectations for graduate students are much higher and therefore the grading is more rigorous. Candidates for any graduate-level degree or certificate must attain a final cumulative grade point average of 3.0 before the degree or certificate will be conferred.

The following letter and special grades are used across all graduate programs. Please note that this document sets the minimum standards for the College. Individual degree programs may have stricter GPA and course grade requirements.

A: "A" indicates outstanding work

B: "B" means that the work is satisfactory

C: "C" (2.0) is deemed unsatisfactory at the graduate level. No more than two courses at the C level (2.0 or higher) will be counted as acceptable toward a graduate degree. Students may be permitted to repeat only two courses and may repeat each course only one time. Those who receive more than two C level grades will be automatically dismissed from their program of study.

Any grade lower than a C - will not be acceptable for graduate-level work and cannot be counted as credit towards the degree. However, the grade will be counted toward the graduate student's GPA. If a student receives a grade lower than a C in a required course, the student will be required to retake the course. A course may only be retaken one time. The most recent grade will replace the first grade and will be factored into the student's GPA.

Please see "Academic Requirements and Policies" in the Graduate Catalog at <https://catalog.merrimack.edu/index.php> for more information.

Student requirements and their corresponding weights are summarized as follows:

Assignment	Weight
Project proposal	10%
Data acquisition and exploration report	10%
Data preprocessing and feature engineering report	10%
Model selection and development report	10%
Model evaluation and interpretation report	20%
Capstone project final report	30%
Peer review	5%
Participation	5%

Each requirement is discussed in the Weekly Assignment flow at the end of the syllabus.

Grading-Related Policies for this Course

All assignments must be submitted through Canvas by 11:59PM of the due date listed in the Weekly Topics & Assignments section and in Canvas. Please note that there will be **NO late work accepted**. Final grades will be based on a percentage of possible points earned, but the instructor reserves the right to adjust the course grade.

1. Academic Integrity

See Merrimack College Student Handbook for college's Academic Integrity Policy. The academic integrity code and policy is also posted in the Graduate Catalog:

<https://catalog.merrimack.edu/content.php?catoid=9&navoid=202#academic-integrity>

Here is a brief excerpt from that: "...Academic integrity is fundamental to creating and maintaining an atmosphere of cooperation and trust. It is thus a concern for everyone in the college community. The academic integrity code below is designed to help students understand what is not permissible in their academic and intellectual lives at the college. It seeks to protect students from unintentional acts of dishonesty and to preserve the trust inherent in the student- teacher relationship, which is compromised if suspicion arises regarding the integrity of a student's work. The code is also designed to inform students of the rules which will be used to judge academic integrity infractions..."

Specifically concerning the production of code it is important to recall that using code, text or other forms of media, such as homework solutions, from previous offerings of the course is forbidden, except as a "reference" or "guide" to work from, and even as such, **it is necessary to explicitly cite all consulted sources**. Tutors or other students who have taken the course in the past may help current students, but should not show them the solutions that they have developed for assignments. The lack of detailed citation of previous offerings of the course or any other similar sources, notably from the Internet or Generative AI systems, will be considered as submission of copied material, and a serious breach of academic integrity. Simply put, any work that is not your own should be clearly identified either through citations in written work or in comments in submitted files. This includes not only things that are directly copied from other's work, but also for ideas and procedures gathered from outside sources. It is not necessary to cite the materials presented in class.

2. Academic Accommodations from the Accessibility Services Office

Merrimack College provides reasonable accommodations for students with documented disabilities. Students who have, or think they may have, a disability are invited to contact the Accessibility Services Office via the online request form found on the Accessibility Services website: www.merrimack.edu/aso, email: accessibilityservices@merrimack.edu, or by visiting us on the third floor of McQuade Library (in the event we are open).

Students are encouraged to contact the office as soon as possible to ensure adequate time to meet and create a plan. Students already registered with Accessibility Services are encouraged semesterly to request for their letters to be emailed and students are responsible to then email the PDF to their instructors personally. Accommodations cannot be made retroactively.

3. Live On-line Meeting Attendance

Participation in weekly live on-line meetings is highly encouraged, but it is not mandatory. All sessions will be recorded, and recordings will be posted; however, when not attending students forego the ability to directly ask questions about topics being discussed, or other parts of the course.

4. Communication

Class communication will come through official college email (merrimack.edu) and Canvas announcements as needed. Those will be the primary methods for dissemination of information between instructors and students, so please check your email at least once a day. Instructors check their email several times a day, so it is the quickest way to reach them with questions. Important announcements will be archived through Canvas. When emailing your instructor, use your Merrimack account, include a greeting with the class code and a closing signature with your name, or they may not respond.

If you are having trouble with an assignment, please do not suffer in silence. Let your instructor know if you are struggling, either by email or by scheduling an appointment. We are all interested in seeing students succeed, and are more than happy to provide extra help.

5. Requests for Extensions

The general policy is that, outside of properly verified serious medical emergencies* (as defined below), extensions are not given, which applies to all assignments. Missing an assignment without an acceptable reason (to be clear, that means a serious medical emergency, as defined below) will result in 0 points for the exam or a project. The intent here is not to penalize anyone – quite to the contrary, it is to create a level playing field so that no one has a unique and an unfair advantage. All assignment due dates are published (see the Live Meetings, Topics & Assignments section below) and will not change, barring a natural or other emergency – please consider those dates when planning any non-class related activities.

**Serious medical emergency is defined as an injury or illness that is acute and poses an immediate risk to a person's life or long term health. To be "properly verified", the said serious medical condition must be attested to by hospitalization and related medical treatment documentation.*

6. Mental Health

It is important to make your mental health a priority! We will do activities in class to support your mental health and wellness, and I also encourage you to engage in your own self-care habits outside of class. If you want more information or resources, please come see me! If you are struggling with your mental health, or you believe a classmate is, please talk to me so I can put you in touch with qualified and caring support to get you back on track and feeling better.

7. Suicide Prevention Lifeline

We can all help prevent suicide. The Lifeline provides 24/7, free and confidential support for people in distress, prevention and crisis resources for you or your loved ones, and best practices for professionals.

http://www.ulifeline.org/stay_well

National Suicide Prevention Lifeline 1-800-273-8255(TALK) or Call or Text 988

8. Counseling Services

Counseling Services are available to Graduate students at the Counseling Center at Merrimack via Uwill for telecounseling services only. Telecounseling includes video, chat, phone, message. Uwill is HIPPA and FERPA compliant which means they follow the strictest privacy guidelines. To access Uwill, register with your Merrimack email address at app.uwill.com. To receive support after hours call 978-837-5444 to be connected to the Uwill 24/7/365 crisis line which offers phone assistance only. The Counseling Center is located on the third floor of the Sakowich Center.

9. Student Success Resources

It takes a village to progress in your knowledge. The numerous resources available at Merrimack College to assist you in meeting your goals are outlined in the **Data Science Hub in Canvas**.

For assistance in navigating the student success resources please reach out to the graduate advising team via ecs-grad-advising@merrimack.edu

Weekly Topics & Assignments

Week 1: Team Structure and Project Proposal

- Select a project topic & crafting a project proposal
- Define objectives, scope, and deliverables

Week 2: Data Acquisition and Finalizing the Proposal

- Source and clean data, finalize the project proposal, & craft an analysis plan

Week 3: Exploratory Data Analysis

- Apply exploratory data analysis (EDA) techniques to generate visualizations and data summary statistics

Week 4: Data Cleaning, Preprocessing and Feature Engineering

- Clean & pre-process (handle missing values, outliers, etc.) data, & transform or scale data
- Select features (variables) and engineer

Week 5: Model Selection and Development

- Select appropriate machine learning algorithms, validate model assumptions, & refine algorithms as needed

Week 6: Model tuning, validation, & exploration

- Cross-validation and hyperparameter optimization, model tuning, and evaluate model metrics
- Final model selection and evaluation

Week 7: Finalizing Model & Interpretation

- Interpret and explain results in the context of the question, hypothesis, & prediction
- Explore results with visual storytelling

Week 8: Final Deliverables

- In-class presentations of capstone projects
- Submit final written deliverables and code, & develop an e-portfolio